

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Dot .Net Core and Progressive Web Application Practical

Practical – VII

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Class: TYIT	Batch: 1
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Part B - Program

A}. Develop a PWA that works in offline mode using Service Worker.

Code:

Manifest.json

```
{
  "name": "College Information Portal",
  "short_name": "CIP",
  "start_url": "/",
  "display": "standalone",
  "background_color": "#ffffff",
  "theme_color": "#0d6efd",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
      "type": "image/png",
      "purpose": "any"
    }
  ]
}
```

Service Worker

```
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open("V1").then(c =>
      c.add("/offline.html"))
  );
});

self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() =>
      caches.match("/offline.html"))
  );
});
```

Layout.cshtml

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>@ViewData["Title"] - WebApplication1</title>
  <script type="importmap"></script>
  <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
  <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
  <link rel="stylesheet" href="~/WebApplication1.styles.css" asp-append-version="true" />
  <link rel="manifest" href="/manifest.json" />
</head>
<body>
  <header>
    <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-bottom box-shadow mb-3">
      <div class="container-fluid">
        <a class="navbar-brand" asp-area="" asp-controller="Home" asp-action="Index">WebApplication1</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target=".navbar-collapse" aria-
controls="navbarSupportedContent"
        aria-expanded="false" aria-label="Toggle navigation">
          <span class="navbar-toggler-icon"></span>
        </button>
        <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
          <ul class="navbar-nav flex-grow-1">
            <li class="nav-item">
              <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-action="Index">Home</a>
            </li>
            <li class="nav-item">
              <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-action="Privacy">Privacy</a>
            </li>
          </ul>
        </div>
      </div>
    </nav>
  </header>
  <div class="container">
    <main role="main" class="pb-3">
      @RenderBody()
    </main>
  </div>

  <footer class="border-top footer text-muted">
    <div class="container">
      &copy; 2026 - WebApplication1 - <a asp-area="" asp-controller="Home" asp-action="Privacy">Privacy</a>
    </div>
  </footer>
  <script src="~/lib/jquery/dist/jquery.min.js"></script>
  <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
  <script src="~/js/site.js" asp-append-version="true"></script>
  @await RenderSectionAsync("Scripts", required: false)
  <script>
    if('serviceWorker' in navigator) {
      navigator.serviceWorker.register('/service-worker.js');
    }
  </script>
</body>
</html>
```

offline.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Offline - SHETH L.U.J. & SIR M.V. College</title>
</head>
<body>
  <center>
    <br><br><br>

    <h1>❏ SHETH L.U.J. & SIR M.V. College</h1>

    <h2>❏ You're Offline</h2>

    <hr width="50%">

    <p>
      It looks like your device is not connected to the internet.
    </p>

    <p>
      Please check your Wi-Fi or mobile data connection and try again.
    </p>

    <p>
      ❏ <b>No Internet Connection</b>
    </p>

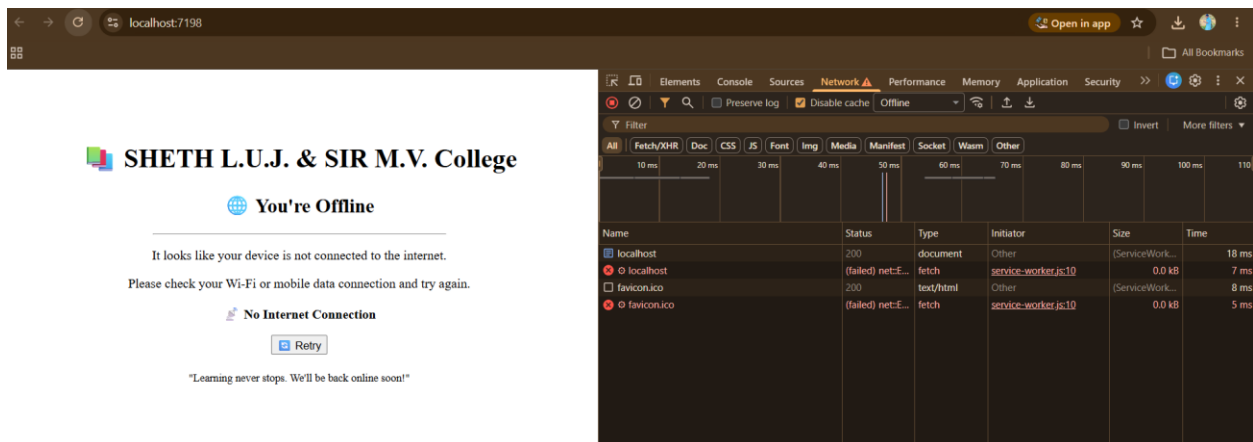
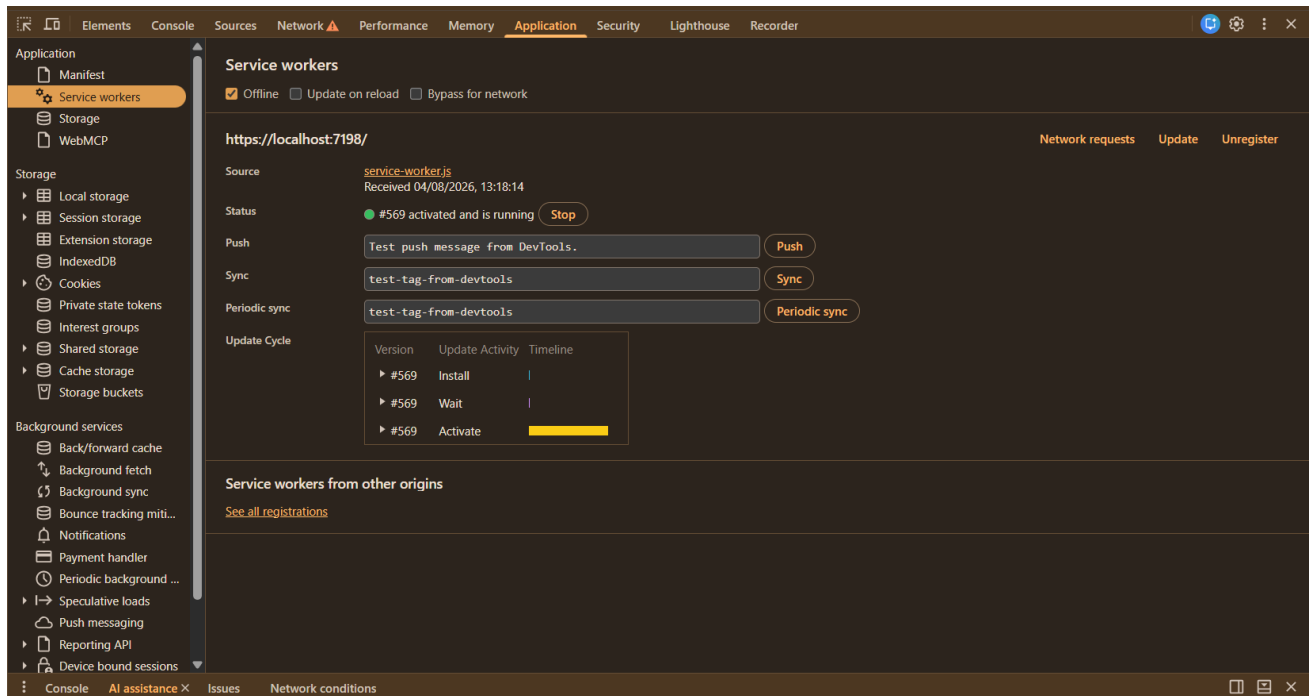
    <button onclick="location.reload()">
      ❏ Retry
    </button>

    <br><br>

    <small>
      "Learning never stops. We'll be back online soon!"
    </small>

  </center>
</body>
</html>
```

Output:



B}. Create a Student Record Management PWA.

Code:

Manifest.json

```
{
  "name": "",
  "short_name": "",
  "start_url": "",
  "display": "",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128x128",
```

```

        "type": "image/png",
        "purpose": "any"
    },
    {
        "src": "icons/icon2.png",
        "sizes": "512x512",
        "type": "image/png",
        "purpose": "any"
    }
]
}

```

service-worker.js

```

const CACHE = "v1";

self.addEventListener("install", e => {
    e.waitUntil(
        caches.open(CACHE).then(cache => {
            return cache.addAll([
                "/",
                "/Home/Index",
                "/css/site.css",
                "/js/site.js"
            ]);
        })
    );
});

self.addEventListener("fetch", e => {
    e.respondWith(
        caches.match(e.request).then(response => {
            return response || fetch(e.request);
        })
    );
});

```

Layout.cshtml

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>@ViewData["Title"] - Practical_7B</title>
    <script type="importmap"></script>
    <link rel="stylesheet" href="~/lib/bootstrap/dist/css/bootstrap.min.css" />
    <link rel="stylesheet" href="~/css/site.css" asp-append-version="true" />
    <link rel="stylesheet" href="~/Practical_7B.styles.css" asp-append-version="true" />
    <link rel="manifest" href="Manifest.json"/>

```

```

</head>
<body>
  <header>
    <nav class="navbar navbar-expand-sm navbar-toggleable-sm navbar-light bg-white border-bottom box-shadow mb-3">
      <div class="container-fluid">
        <a class="navbar-brand" asp-area="" asp-controller="Home" asp-action="Index">Practical_6A_B</a>
        <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target=".navbar-collapse" aria-controls="navbarSupportedContent"
          aria-expanded="false" aria-label="Toggle navigation">
          <span class="navbar-toggler-icon"></span>
        </button>
        <div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
          <ul class="navbar-nav flex-grow-1">
            <li class="nav-item">
              <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-action="Index">Home</a>
            </li>
            <li class="nav-item">
              <a class="nav-link text-dark" asp-area="" asp-controller="Home" asp-
action="Privacy">Privacy</a>
            </li>
          </ul>
        </div>
      </div>
    </nav>
  </header>
  <div class="container">
    <main role="main" class="pb-3">
      @RenderBody()
    </main>
  </div>

  <footer class="border-top footer text-muted">
    <div class="container">
      &copy; 2026 - Practical_7B- <a asp-area="" asp-controller="Home" asp-action="Privacy">Privacy</a>
    </div>
  </footer>
  <script src="~/lib/jquery/dist/jquery.min.js"></script>
  <script src="~/lib/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
  <script src="~/js/site.js" asp-append-version="true"></script>
  @await RenderSectionAsync("Scripts", required: false)
  <script>
    if('serviceWorker' in navigator)
    {
      navigator.serviceWorker.register('/service-worker.js');
    }
  </script>
</body>
</html>

```

Student.cs

```
namespace Practical_6A_B.Models
{
    public class Students
    {
        public class Student
        {
            public int RollNo { get; set; }
            public string Name { get; set; }
        }
    }
}
```

HomeController.cs

```
using System.Diagnostics;
using Microsoft.AspNetCore.Mvc;
using Practical_6A_B.Models;
using static Practical_6A_B.Models.Students;

public class HomeController : Controller
{
    public IActionResult Index()
    {
        Student S = new Student();
        S.RollNo = T008;
        S.Name = "Aman ";

        return View(S);
    }
}
```

Index.cshtml

```
@{
    ViewData["Title"] = "Home Page";
}

<div class="text-center">
    <h1 class="display-4">Welcome</h1>
    <p>Learn about <a href="https://learn.microsoft.com/aspnet/core">building Web apps with ASP.NET
Core</a>.</p>
    <h1>Offline Page</h1>
    <h2>Student Record</h2>
    <p>Roll No : @Model.RollNo</p>

    <p>Name : @Model.Name</p>
</div>
```

Output:

